

Vale Arden

Principal Fictional Engine Architect

Maximum-feature synthetic CV intended as a final integration test for the Lua-driven CV generator. It exercises scalar fields, arrays, nested tables, functions, booleans, numbers, custom sections, repeated project cards, education cards, tags, and dense text.

Loc: Aster Test Region
env: Linux / Wayland / Vulkan
mail: vale.arden@example.test
git: github.com/fake-vale-arden

/// SKILLS

Build:

CMake Ninja Git CI

Debug:

RenderDoc Validation Layers Sanitizers
Profiling

Engine:

ECS Render Graph RHI Asset Pipeline
Editor

Graphics:

Vulkan OpenGL GLSL SPIR-V PBR
Deferred Rendering

Languages:

C++ C Lua Python Bash

/// METRICS

AutomatedChecks: 312
RenderPasses: 27
Reproducible: true
ShaderPermutations: 64
SyntheticProjects: 4
UsesNetwork: false

/// PRINCIPLES

- > Make ownership explicit
- > Keep GPU state inspectable
- > Prefer data-driven pass declarations
- > Treat validation output as an engineering signal
- > Optimize only after measuring

/// EDUCATION

MSc Rendering Systems

Fictional Institute of Advanced Engine Systems
Status: Completed

BSc Computer Science

Imaginary University
Status: Completed

/// AVAILABILITY

Hybrid: true
OnSite: true
Remote: true
Travel: false

/// PROJECTS

Helio Engine (Architect)

Synthetic engine integrating a renderer, editor, asset pipeline, scene system, and profiling tools behind a native C++ codebase.

C++ Vulkan ECS ImGui CMake

GraphPilot (Graphics Architect)

Fictional render graph that infers pass dependencies from resource declarations and emits an executable frame schedule.

C++ Vulkan Render Graph Synchronization

Material Atlas (Rendering Engineer)

Synthetic bindless material system supporting texture indexing, material buffers, and shader-side lookup.

Vulkan GLSL Descriptor Indexing SPIR-V

Viewport Forge (Editor Engineer)

Fictional editor viewport with camera controls, scene selection, transform gizmos, overlays, and render diagnostics.

C++ ImGui ImGuizmo Vulkan

/// ARCHITECTURE

Memory:

Allocation: Suballocated GPU resources
Lifetime: Frame-local Scene-local Global
Staging: Persistent upload arena

Pipeline:

AssetImport: Mesh Material Texture Metadata
Runtime: Load Cache Submit
Shader: Compile Reflect Validate

Renderer:

API: Vulkan 1.3
Bindless: true
DynamicRendering: true
Reflection: true
Style: Explicit